

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th May 2023

Subject CP3 – Communication Practice

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Mark allocations are shown in brackets.*
- 2. Attempt all questions beginning your answer to each question on a new page.*
- 3. Attempt all sub-parts of the question in one document only, unless otherwise instructed to do so.*
- 4. Do save your work in solution template on a regular basis.*
- 5. All the detailed guidelines are available on exam screen.*
- 6. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

You are an Actuary working as the head of Risk management function of a Life insurance company which specialises in selling annuities.

Interest rates have been increasing in most countries and has increased in your country by 150 bps in the span of a year. There is a meeting of the Management team on the Strategy of the Company where among other things the likely impact of increasing interest rates on the Company's capital and business plans will be discussed. There is a new Head of sales who has had most of his experience on the sales side of a Credit Card Company. He has read some news articles excerpts of which are given below.

Central banks around the world have been raising the cost of borrowing to try to dampen down rising prices. After years of low interest rates, that has come as a shock. Banks holding Government bonds, that go down in value when interest rates rise, have suddenly found their assets are worth less. That has affected the whole banking sector, but smaller banks are more vulnerable.

Bank ABC which shut down recently had in its quest for higher investment returns dramatically increased its holdings of long-term securities accounting for them on a hold-to-maturity basis. The market value of these bonds decreased significantly when the Central Bank of the country started increasing interest rates to bring down inflation, causing unrealized losses on the portfolio. Higher interest rates also raised borrowing costs throughout the economy and some clients started pulling money out to meet their liquidity needs.

In Country B, many pension schemes struggled to find the required cash in a short timescale (even highly liquid assets such as equities took several days to sell) and this also meant that many had to sell bonds, thereby reducing their value further.

He has written to you as follows:

Dear Actuary,

"I have attached snapshots of some news articles I read. In my induction I was briefly taken through our investment strategy, and I understood that we invest in long term bonds and have exposure to derivatives. You may be aware that the demand for annuity products is very high now, due to increased awareness of annuities and high interest rates. I am sure we would be actively monitoring risks, especially in the current environment. Before the upcoming meeting on Strategy, I would like to get an overview of the risk management framework. I have some questions which come to my mind after reading about what happened with Banks and pension schemes abroad.

Will we have similar risks as the banks had? Is it possible for our existing customers to withdraw their annuity policies? Would there be a problem if there was a spike in withdrawals? Whether there would be any impact on our profits because of losses on bonds?

I am concerned that we may have to reverse our strategy if our ability to write business is under threat due to the external environment. Please drop me a small note giving me an overview of our risk management framework and your assessment of the current situation in as simple a manner as possible.

Regards

Ram

The risk management function monitors all the key parameters on an ongoing basis every quarter. However, due to the sudden increase in interest rates, the frequency of monitoring has been stepped up and some tactical calls have been taken. The excerpts of the note which was prepared is provided in the Annexure. You may use the information provided in the note to answer the questions.

- Q. 1)** Draft an email response to the Head of Sales explaining in brief the risk management framework, how risks are being monitored, the current capital position of the Company and continuity of business plans. **[90]**
- Q. 2)** Explain how you decided on
- i)** Structure of your response (3)
 - ii)** The depth of your explanation (5)
 - iii)** Filtering information (2)
- [10]**

Annexure

Pre read for the meeting of the Risk Committee

The main type of annuity which is sold by the Company is a **Regular premium deferred annuity**. Premiums are paid for one to five years and annuity payments start with a deferment period of one year or five years and are payable for life of the annuitant. During the deferment period surrenders are allowed but with heavy surrender penalties.

Pricing

Pricing is reviewed on a weekly basis. The Company uses hedging to lock into interest rates for cashflows that occur in future. Therefore, yields expected on the future premiums are close to current yields.

Surrender penalties

Surrender values factors are derived based on the yields at the time of pricing plus 200 basis points.

Investment strategy and risk management

There are two funds the **annuity fund** and the **surplus fund**.

The investment strategy for the annuity fund is to maximise yields subject to duration matching.

The Strategic Asset Allocation for the annuity fund requires investment in long duration GSecs, Corporate bonds and cash or cash equivalents of at least 2% of the fund.

The Strategic Asset Allocation for the surplus fund requires investment in short duration GSecs and short duration Corporate Bonds.

The intent of the Company's derivative strategy is to manage interest rate risk and duration matching.

Interest rate risk arises due to the assumption at the time of pricing of the ability of the Company to lock into yields on positive cashflows which will occur in the future. Positive cashflows arise on account of premiums payable in future, coupons and redemptions from investments which have already been made. A fall in interest rates would mean lower interest earnings for the Company exposing the Company to interest rate risk. The Company takes exposure to interest rate derivatives to lock into interest rates to the extent of 90% of these cashflows which occur in future. The Company monitors its duration mismatch by taking into account the derivative exposure. The intention is to minimize the gap to less than 0.5 years.

While the Company uses derivatives for hedging purposes with the objective of managing its interest rate risk, the Company recognizes that the key additional risk that it is exposed to is liquidity risk.

The Company may face liquidity risk due to markets becoming illiquid. The risk is managed by way of option available to the Company to terminate the contract as and when desired. Liquidity risk may also arise on account of periodic margin calls being made by the respective counterparties where the mark to market loss on the derivative contract exceeds the pre-defined thresholds requiring the Company to pay the differential amounts. The company mitigates this risk by holding liquid investments like Government securities and cash.

Liquidity risk may also arise when policyholders surrender their policies due to rising interest rates. If interest rates rise too rapidly, then policyholders may surrender policies faster than expected, potentially resulting in cash flow obligations that exceed returns on investment assets.

The Company monitors the actual cash and the liquidity of the securities it holds. It is expected that Central Government Securities are realizable with minimal loss.

Stress tests

The Company periodically performs stress tests to see the impact on solvency and liquidity. The prescribed stress test is a 200 bps increase and decrease in interest rate. Surrenders are stressed by 20% from the current best estimate. Due to the duration matching strategy followed by the Company, the solvency is well protected. The surplus assets provide some sensitivity to the solvency position as any changes in the value of the surplus assets do not have corresponding changes in the liability. In an interest rate up the total assets fall in value more than that of liability which impacts the solvency slightly.

Liquidity Coverage ratio is covered under Base scenario and a scenario in which interest rates increase by 100 bps along with spike in surrenders by 50%. The liquidity coverage ratio which is the ratio of the available cash to the required cash is quite comfortable and exceeds the thresholds set.

However, in view of the prevailing scenario of sudden jump in interest rates the Company performed additional scenarios of interest rate movements and liquidity stress tests.

Scenario 1

The current best estimate of surrenders is very low at 2%. The surrender penalties provide a cushion to absorb any losses that would arise on account of assets and derivatives having

lower values as compared to when they were bought. The scenario tested was a 100% increase in surrenders across all cohorts from the current rates plus a 200 bps increase in interest rates.

Scenario 2

A more severe scenario of increase in interest rates from the current levels by 200 bps along with an increase in base rates tested in Scenario 1 plus 100% surrenders from those cohorts where exiting customers even after paying surrender penalties stood to gain by way of higher annuities. In such a scenario the surrender rate approximately translated to 10% of the book. In such a scenario, the Company would suffer a loss as it would have to redeem assets to pay the outflows as the surrender penalty is not expected to cover the loss from the sale of the asset. Additional losses would occur as there would be a need for unwinding some of the derivative trades as in the event of surrender these cashflows would not need to be hedged. The losses are significant and solvency falls to 120%.

Liquidity coverage ratio however continues to be high even if the increase in interest rate were to be accompanied with high surrenders. The results of stress tests are provided below.

Parameter	Current position	200 bps further increase no change in surrender pattern	Scenario 1	Scenario 2
Solvency	150%	135%	130%	120%
Liquidity coverage ratio	400%	350%	325%	300%
Projected solvency allowing for one year's new business	151%	134%	131%	125%

As per the Company's policy on Capital, the level of solvency should be above 135%. In case in any of the scenarios the Company sees that the Solvency level falls below 120%, it would have to put in measures to bring it back up or strengthen capital position. The Regulatory limit is 100%.

In case the interest rates were to increase from the current high levels by a further 200 bps with no change in surrender pattern then the solvency is at 135%. If this was accompanied with increased surrenders solvency is at 130%. However, if surrenders spike significantly then the solvency falls to a level at which the Company would need to trigger capital raising as per the internal guidelines. The Company is monitoring the situation.

The Risk Committee has decided on the following:

- Continue to monitor surrenders and interest rates
- Deviate from the Strategic Asset Allocation to increase cash holdings temporarily so as to avoid forced redemptions
- For new business enter into derivatives to a lesser extent of 80%
- No change in business plan
